

Flow Matching with Stochastic Interpolants: Three Dirac Masses and Gaussian Endpoint Geometry

Abstract

We study the probability-flow velocity associated with stochastic interpolants of the form $X_t = a(t)X_0 + b(t)X_1$. The first case transports a standard Gaussian source toward a symmetric mixture of three Dirac masses. In this semi-discrete setting the density of X_t is an explicit Gaussian mixture, so the flow-matching velocity is available as a closed-form conditional expectation. Numerical experiments compare linear, variance-preserving, and cosine schedules and contrast their curved probability-flow trajectories with the straight rays of semi-discrete optimal transport. The second case takes Gaussian endpoints. We derive the covariance path, the linear velocity field, and a closed-form flow map for arbitrary schedules a, b . The terminal map matches the target covariance but agrees with the Gaussian optimal-transport map only when the endpoint covariances commute.

1 Introduction

Flow matching learns deterministic velocity fields whose continuity equation transports a simple base distribution to a data distribution. Recent formulations by Lipman et al. [2] and the stochastic-interpolant framework of Albergo et al. [1] show that many generative modeling objectives can be obtained by specifying an interpolation X_t between two endpoint random variables and regressing the conditional mean velocity. Rectified flow [3] emphasizes the same idea from the perspective of learning straight or progressively straightened flows.

The construction is closely related to, but distinct from, optimal transport (OT). The OT map is selected by a global variational principle and, for quadratic cost, is the gradient of a convex potential [4–6]. A stochastic interpolant instead starts with a coupling of (X_0, X_1) , a time schedule, and a conditional expectation. The resulting flow has the prescribed marginals, but it need not be the Brenier flow. This note makes that distinction explicit in two examples where all formulas can be written down and checked numerically.

Contributions. First, for $X_0 \sim \mathcal{N}(0, I_2)$ and X_1 supported on three Dirac masses, we derive the exact mixture density and the exact conditional expectation velocity, then integrate the probability-flow ODE in PyTorch. Second, we compare the resulting trajectories with the semi-discrete OT rays. Third, for independent Gaussian endpoints, we compute the covariance path, velocity matrix, and a closed-form flow map. We prove that the terminal map equals the Gaussian OT map if and only if the two covariance matrices commute.

2 Stochastic Interpolants and Conditional Velocities

Let $X_0 \sim \rho_0$, $X_1 \sim \rho_1$, with a chosen coupling, and let

$$X_t = a(t)X_0 + b(t)X_1, \quad t \in [0, 1],$$

where a, b are continuous on $[0, 1]$, differentiable on $(0, 1)$, and satisfy

$$a(0) = 1, \quad b(0) = 0, \quad a(1) = 0, \quad b(1) = 1.$$

Define $\rho_t = \text{Law}(X_t)$. Differentiating the interpolant gives

$$\dot{X}_t = \dot{a}(t)X_0 + \dot{b}(t)X_1.$$

The associated flow-matching velocity is

$$u_t(x) = \mathbb{E}[\dot{a}(t)X_0 + \dot{b}(t)X_1 \mid X_t = x]. \quad (1)$$

Under standard regularity assumptions this velocity solves the continuity equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t u_t) = 0.$$

The deterministic probability-flow ODE

$$\frac{d}{dt}x_t = u_t(x_t)$$

therefore transports ρ_0 along the same one-time marginals on the open time interval, with endpoint laws understood by limit when the velocity is singular at 0 or 1. The schedule changes the velocity even when the endpoint coupling is fixed.

The experiments use three schedules:

Name	$a(t)$	$b(t)$
linear	$1 - t$	t
variance preserving	$\sqrt{1 - t}$	$\sqrt{t}$
cosine	$\cos(\pi t/2)$	$\sin(\pi t/2)$

The square-root schedule has singular endpoint derivatives; in the numerical ODE integration we use the interval $[10^{-4}, 1 - 10^{-4}]$.

3 Three Dirac Masses

Let $X_0 \sim \mathcal{N}(0, I_2)$, and let X_1 be uniform on three points y_1, y_2, y_3 forming an equilateral triangle centered at the origin. In this section X_0 and X_1 are independent. The notebook `python/flow_matching_dirac_mixture.ipynb` implements the following closed-form formulas with PyTorch.

3.1 Density and velocity

Conditionally on $X_1 = y_k$,

$$X_t \mid X_1 = y_k \sim \mathcal{N}\left(b(t)y_k, a(t)^2 I_2\right).$$

With weights $\pi_k = 1/3$, the density is

$$p_t(x) = \sum_{k=1}^3 \pi_k \varphi_{a(t)^2 I_2}(x - b(t)y_k). \quad (2)$$

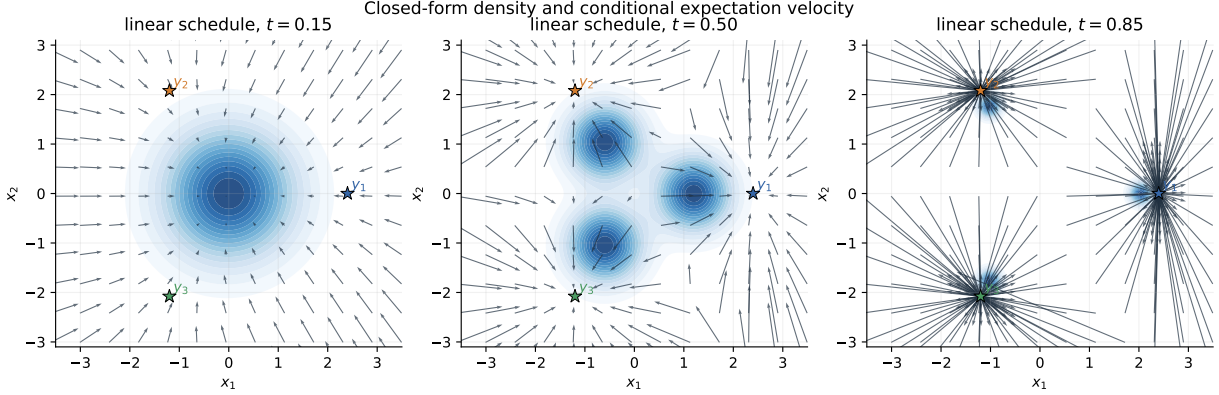


Figure 1: Exact density contours and conditional expectation velocity for the linear schedule. The density is the closed-form mixture (2); arrows show the PyTorch evaluation of (4).

The posterior probability of the k -th endpoint is

$$\gamma_k(x, t) = \frac{\pi_k \exp(-\|x - b(t)y_k\|^2 / (2a(t)^2))}{\sum_{\ell=1}^3 \pi_\ell \exp(-\|x - b(t)y_\ell\|^2 / (2a(t)^2))}. \quad (3)$$

Since $X_0 = (X_t - b(t)X_1)/a(t)$ for $a(t) > 0$, substituting (3) into (1) gives

$$u_t(x) = \frac{\dot{a}(t)}{a(t)}x + \left(\dot{b}(t) - \frac{\dot{a}(t)b(t)}{a(t)} \right) \sum_{k=1}^3 \gamma_k(x, t)y_k. \quad (4)$$

3.2 Probability-flow trajectories

Figure 2 shows trajectories of the probability-flow ODE. Each curve is colored by the nearest Dirac mass at the final numerical time. The schedules share the same endpoint constraints but visibly change the geometry of the paths. Linear interpolation removes source variance quickly, while the variance-preserving and cosine schedules keep broader intermediate clouds and produce different bending before the terminal collapse.

3.3 Comparison with semi-discrete optimal transport

Because the target points form an equilateral triangle and the source Gaussian is rotationally invariant, the equal-mass semi-discrete OT Laguerre cells have equal dual weights. Since the three targets have the same norm, those cells are exactly the nearest-vertex cones. The OT map is therefore

$$T_{\text{OT}}(x) = y_{\arg \min_k \|x - y_k\|},$$

up to cell boundaries of zero Gaussian measure. Its displacement interpolation follows straight rays,

$$x_t = (1 - t)x + tT_{\text{OT}}(x).$$

Figure 3 contrasts these rays with the conditional-expectation flows. Both constructions connect the same endpoint measures, but the stochastic-interpolant flow is selected by the coupling and schedule, not by the quadratic transport cost.

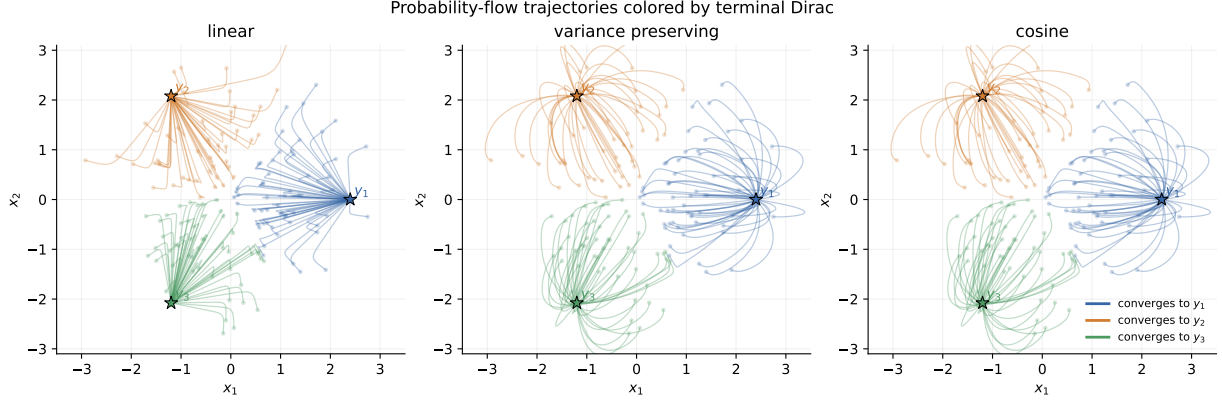


Figure 2: Sample trajectories of the flow-matching ODE for three schedules. Colors indicate the Dirac mass to which the final point is closest.

4 Gaussian Endpoint Geometry

We now take independent Gaussian endpoints

$$X_0 \sim \mathcal{N}(0, \Sigma_0), \quad X_1 \sim \mathcal{N}(0, \Sigma_1),$$

where Σ_0, Σ_1 are symmetric positive definite matrices.

Theorem 1 (Gaussian stochastic-interpolant flow). *Let $X_t = a(t)X_0 + b(t)X_1$ with independent centered Gaussian endpoints. Define*

$$\Sigma_t = a(t)^2 \Sigma_0 + b(t)^2 \Sigma_1, \quad A = \Sigma_0^{-1/2} \Sigma_1 \Sigma_0^{-1/2},$$

and

$$M_t = a(t)^2 I + b(t)^2 A.$$

Then $X_t \sim \mathcal{N}(0, \Sigma_t)$, and the conditional velocity is linear,

$$u_t(x) = B_t x, \quad B_t = (a(t)\dot{a}(t)\Sigma_0 + b(t)\dot{b}(t)\Sigma_1)\Sigma_t^{-1}. \quad (5)$$

The flow map from time 0 to time t is

$$T_t = \Sigma_0^{1/2} M_t^{1/2} \Sigma_0^{-1/2}. \quad (6)$$

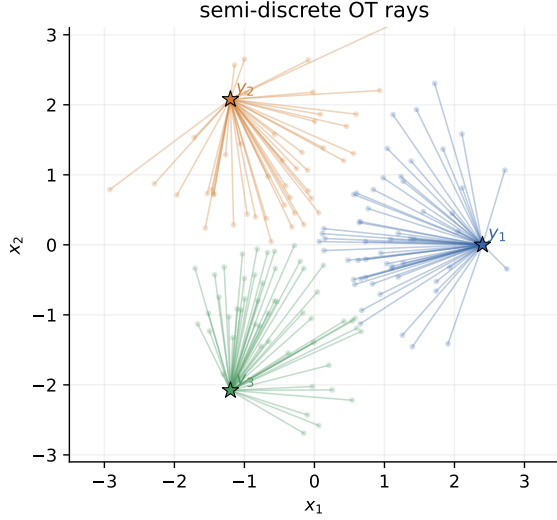
In particular, if $a(1) = 0$ and $b(1) = 1$,

$$T_1 = \Sigma_0^{1/2} \left(\Sigma_0^{-1/2} \Sigma_1 \Sigma_0^{-1/2} \right)^{1/2} \Sigma_0^{-1/2}. \quad (7)$$

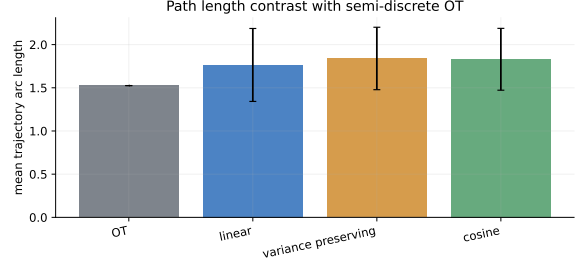
The Gaussian optimal-transport map is

$$T_{\text{OT}} = \Sigma_0^{-1/2} \left(\Sigma_0^{1/2} \Sigma_1 \Sigma_0^{1/2} \right)^{1/2} \Sigma_0^{-1/2}. \quad (8)$$

Moreover, $T_1 = T_{\text{OT}}$ if and only if $\Sigma_0 \Sigma_1 = \Sigma_1 \Sigma_0$.



(a) Semi-discrete OT rays.



(b) Mean trajectory arc length.

Figure 3: The symmetric OT map partitions the source by nearest target point and moves along straight segments. The flow-matching trajectories are generally curved and have schedule-dependent arc length.

Proof. The covariance formula follows from independence:

$$\text{Cov}(X_t) = a(t)^2 \Sigma_0 + b(t)^2 \Sigma_1.$$

For a joint centered Gaussian vector, conditional expectation is linear. Since

$$\text{Cov}(X_0, X_t) = a(t) \Sigma_0, \quad \text{Cov}(X_1, X_t) = b(t) \Sigma_1,$$

we have

$$\mathbb{E}[X_0 | X_t = x] = a(t) \Sigma_0 \Sigma_t^{-1} x, \quad \mathbb{E}[X_1 | X_t = x] = b(t) \Sigma_1 \Sigma_t^{-1} x.$$

Substituting into (1) gives (5).

It remains to verify the flow map. Write

$$\Sigma_t = \Sigma_0^{1/2} M_t \Sigma_0^{1/2},$$

where $M_t = a(t)^2 I + b(t)^2 A$. Since M_t and $\dot{M}_t = 2a(t)\dot{a}(t)I + 2b(t)\dot{b}(t)A$ are polynomials in A , they commute. Therefore

$$\frac{d}{dt} M_t^{1/2} = \frac{1}{2} \dot{M}_t M_t^{-1/2}.$$

Differentiating (6) yields

$$\dot{T}_t T_t^{-1} = \Sigma_0^{1/2} \frac{1}{2} \dot{M}_t M_t^{-1} \Sigma_0^{-1/2} = (a\dot{a} \Sigma_0 + b\dot{b} \Sigma_1) \Sigma_t^{-1} = B_t.$$

Also $T_0 = I$, so T_t is the linear flow generated by B_t . The endpoint formula (7) follows by setting $t = 1$.

Formula (8) is the classical Brenier map between centered Gaussians. It is symmetric positive definite. The map (7) also sends Σ_0 to Σ_1 , but it is the Brenier map exactly when it is symmetric.

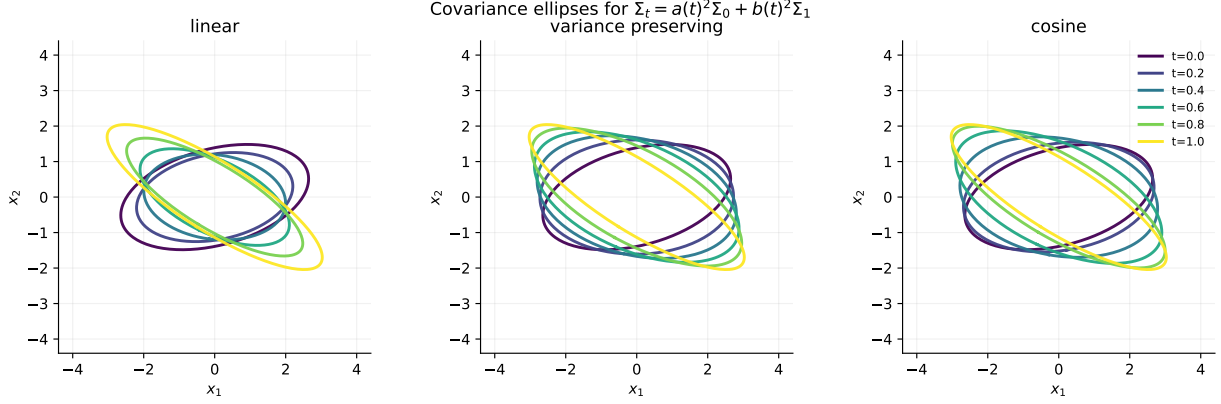


Figure 4: Covariance ellipses for the Gaussian stochastic interpolant under the three schedules. The endpoint ellipses are common; the intermediate covariance geometry is schedule-dependent.

Symmetry of $T_1 = \Sigma_0^{1/2} A^{1/2} \Sigma_0^{-1/2}$ is equivalent to $\Sigma_0 A^{1/2} = A^{1/2} \Sigma_0$, hence to $\Sigma_0 A = A \Sigma_0$. In an eigenbasis of Σ_0 , this condition is equivalent to Σ_1 being block diagonal on the eigenspaces of Σ_0 , which is exactly $\Sigma_0 \Sigma_1 = \Sigma_1 \Sigma_0$. In that commuting case the matrices are simultaneously diagonalizable and the two maps both reduce to multiplication by $\text{diag}(\sqrt{\lambda_{1,i}/\lambda_{0,i}})$. $\square$

4.1 Numerical covariance display

The notebook `python/gaussian_covariance_ellipses.ipynb` plots the ellipses of Σ_t for a non-commuting pair (Σ_0, Σ_1) . Figure 4 shows that the choice of a, b changes the covariance path, even though every schedule has the same endpoints. Figure 5 displays the endpoint map from (7) and the Brenier map from (8); both match the target covariance, but the maps are different because the covariances do not commute.

5 Discussion

These examples separate three objects that are sometimes conflated: the marginal path ρ_t , the deterministic flow that realizes that marginal path, and the OT map selected by the quadratic cost. A stochastic interpolant fixes a coupling and schedule, then obtains a velocity by conditional expectation. This velocity is easy to compute in the examples above, and it is the target of the flow-matching regression in data-driven settings. However, unless the coupling and geometry are specially aligned, the terminal probability-flow map is not the Brenier OT map.

The Gaussian theorem makes this conclusion algebraic. Even when both endpoint distributions are Gaussian and the full flow is linear, the independent endpoint interpolant produces $T_1 = \Sigma_0^{1/2} (\Sigma_0^{-1/2} \Sigma_1 \Sigma_0^{-1/2})^{1/2} \Sigma_0^{-1/2}$, whereas OT requires the symmetric positive map. Equality occurs exactly in the commuting case, where the covariance matrices share eigenspaces and all relevant matrices are simultaneously diagonal.

References

- [1] Michael S. Albergo, Nicholas M. Boffi, and Eric Vanden-Eijnden. Stochastic interpolants: A unifying framework for flows and diffusions. *arXiv preprint arXiv:2303.08797*, 2023.

Endpoint maps both match covariance, but differ as maps

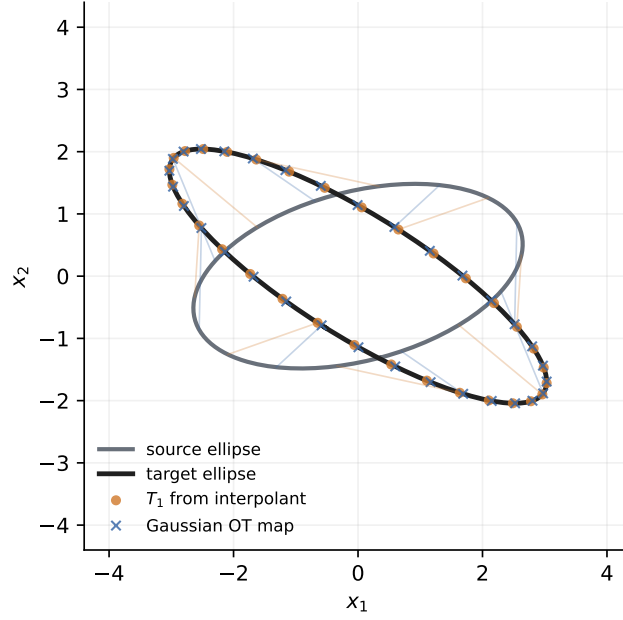


Figure 5: For noncommuting covariances, the stochastic-interpolant endpoint map T_1 and the Gaussian OT map both push Σ_0 to Σ_1 , but they differ as linear maps.

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